Primary Liver Cancer Early Screening Based on Gradient Boosting Decision Tree and Support Vector Machine

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***Abstract*—Primary liver cancer has no obvious clinical symptoms and no effective screening method in the early stage, which leads that its first diagnosis is generally late. Early detection is considered to be the main measure to improve this situation. An early screening method utilizing clinical laboratory dataset was proposed. Firstly, gradient boosting decision tree (GBDT) was used for feature selection, and then two classification methods, support vector machine (SVM) and GBDT, were used for training and testing. The results show that the Kappa index reaches the almost perfect level and the accuracy is over 90%. This method can assist doctors to screen primary liver cancer early.**

Keywords—primary liver cancer; classification; GBDT; SVM; feature selection

I. INTRODUCTION

Primary liver cancer (PLC) is a kind of disease with a high mortality rate in the world, and its causes are complicated with many factors, such as the environment and diet. According to a study in Taiwan [1], after selecting patients based on risk scores, the mortality of the monitored group is reduced compared with the control group and the general population. And the International Union Against Cancer [2] says that if cancer is detected early and treated adequately, one-third of patients can be prevented, and another one-third can be completely cured, but unfortunately 80% of patients missed the best time of diagnosis and treatment. Therefore, the best way to overcome liver cancer is early screening and prevention.

In recent years, with the development of artificial intelligence and the accumulation of medical data, some methods for early prediction of primary liver cancer have proposed. Medical images [3-5] are often used in the identification and prediction of liver cancer to help doctors make decisions. This method first obtained the possible damaged area in the image, then extracted important features, and finally classified. In the field of the clinical laboratory, analyzing the hospital's electronic medical records [6] and the database of bacterial fingerprints established based on clinical data, reduce identification time of bacteria from 24 hours to 30 minutes, which has provided good conditions for timely treatment of patients. Therefore, the performance of blood-based cancer screening is not as good as one based on medical images, but advantages of patient-friendly, easy to sampling, cheap and easily combines with algorithm make it still research focus in the field of cancer screening area.

Data-based machine learning, an important aspect of modern smart technology, is essentially an approximation of the model of the actual problem, beginning with observing the sample and then seeking the rules to predict unknown data. Since support vector machines (SVM) and gradient boosting decision trees (GBDT) were introduced in this field, they excellent learning performance has attracted the focus of scholars and successfully have been applied to intelligent transportation [7], medical diagnosis [4], risk assessment [8], and so on.

SVM and GBDT were used to establish PLC early screening model which based on clinical laboratory dataset. Firstly, the GBDT was used to select features of preprocessed dataset. Then the classifiers were training and testing all features or selected features separately. The classifier used area under curve (AUC) as a parameter optimization measure in Random Searching [9] to train the model. Finally, the performance of above models was evaluated. The method reduces the number of test indices while keeping test accuracy, which makes early successful diagnosis of liver cancer easy.

II. METHODOLOGY

A. Dataset

Experimental data mainly come from Eastern Hepatobiliary Surgery Hospital which has 1069 cases of sample and has been divided into three categories, including health, benign lesions and malignant tumors. The feature dimension of dataset is 78 to 140. Basic features include the following indicators: personal information, blood, urine and stool routine, biochemical markers of liver and renal function, glycolipid metabolism, tumor markers, specific protein, infection marker and clotting module.

B. Preprocessing

In order to ensure the accuracy of classification algorithm, the obviously incomplete features and cases were removed from dataset. The feature dimension of this 1047 case samples

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was reduced to 30. And the samples of health, benign lesions and malignant tumors are 288, 304 and 455 cases respectively. Moreover, some data are not meet the requirements of the algorithm, mainly including range data, text or symbol data and null data. The method of handling those data are the following processes:

- Data normalization: Divided by the maximum value of the feature to make the data between 0 and 1.
- Range data: Remove the symbol, such as taking '<123' as '123'.
- Text or symbol data: Change to numerical value according to the meaning of the data, such as taking '-' as 0, and taking '+' as 1.
- Null data: Fill with 0.

C. Feature Selection

It is obvious that not all features are helpful for performance in the field of pattern recognition. Feature selection can reduce the running time as well as improve computational efficiency and enhance generalization capability. So, it is necessary to find features with closely relevant to classification performance before applying classification algorithm. Tree-based ensemble learning algorithm not only has high prediction accuracy but also can output the features relative importance for the model, that allows us to understand which features are critical in the current model.

GBDT, which consists of many decision trees similar to CART [10], is used in data classification and regression. To regression work, the algorithm [11] was used to select features in this paper. Comparing with traditional boosting algorithm, gradient boosting constructs new model based on reducing the residual of previous model. The main principle is to find a model which makes the residual between sample predictive value and the real value smaller as far as possible in each iteration process. Assuming that the training dataset is $S=\{(x_1,y_1),(x_2,y_2),\cdots,(x_k,y_k)\}$, where $x_i \in R^n$ is the i-th sample vectors, $y_i \in \{+1,-1\}$ is a sample type, (x_i,y_i) is the sample point, and i belongs to 1 to k. Making the maximum number of iterations is T, the loss function is L, the residual is c, and the learner is $F(x)$. The main steps are as follows:

1) Initialization learner:

$$F_0(x)=\arg\min_c \sum_{i=1}^{k} L(y_i,c) \tag{1}$$

2) Iteration process:

a) Calculating the negative gradient r_{ti}:

$$r_{ti}=-\left[\frac{\partial L(y,F(x_i))}{\partial F(x_i)}\right]_{F(x_i)=F_{t-1}(x)} \tag{2}$$

b) Calculating the best fit value c_{tj}: using (x_i,r_{ti}) fits the regression tree to obtain the t-th regression tree. The corresponding leaf nodes are R_{tj}, where j belongs to 1 to J, and J is the number of the regression tree leaf nodes. Shown in Equation (3):

$$c_{tj}=\arg\min_c \sum_{x_i\in R_{tj}} L(y_i,F_{t-1}(x_i)+c) \tag{3}$$

c) Updating learner $F(x)$:

$$F_t(x)=F_{t-1}(x)+\sum_{j=1}^{J} c_{tj} \tag{4}$$

3) Obtaining learner: repeat step 2 until the number of iterations T is satisfied, and the learner is obtained in (5):

$$F(x)=F_T(x)+\sum_{t=1}^{T}\sum_{j=1}^{J} c_{tj} \tag{5}$$

D. Classifiers

1) GBDT

The classification algorithm of GBDT is consistent with the above mentioned regression algorithm except that using linear search methods to calculate the optimal residual fit of negative gradient and leaf nodes. The sample labels are not continuous output value, but discrete categories, so the method uses log-likelihood loss function to fit the difference between predictive and real value. Formulas (2) and (3) are converted to (6) and (7).

$$r_{ti}=-\left[\frac{\partial L(y,F(x_i))}{\partial F(x_i)}\right]_{F(x_i)=F_{t-1}(x)}=\frac{y_i}{1+\exp(yF(x_i))} \tag{6}$$

$$c_{tj}=\arg\min_c \sum_{x_i\in R_{tj}} \log(1+\exp(y_i(F_{t-1}(x_i)+c)))\approx\frac{\sum_{x\in R_{tj}} r_{ti}}{\sum_{x\in R_{tj}} |r_{ti}|(2-|r_{ti}|)} \tag{7}$$

2) SVM

SVM [12] has attracted many scholars in field of machine leaning since it introduced by Vapnik. It has good ability of generalization and classification while has perfect performance about solve pattern recognition problem. At present, it is applied in many fields, such as object detection, medical diagnosis and project prediction. The aim of SVM is to find a hyperplane with largest classification interval for both types of samples.

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The above dataset, for example, through SVM to establish a decision surface. That equivalent to seek the formulas in (8), where ω is hyperplane normal vector, b is hyperplane constant, C is the penalty coefficient, and ξ_i is the errors. Finding partial derivatives for w,b,ξ respectively, after α is introduced as Lagrange multiplier. The equation (8) is converted to (9).

$$\min \frac{1}{2}\|w\|^2 + \mathrm{C}\sum_{i=1}^{k}\xi_i \quad s.t.\begin{cases} y_i(w^{\mathrm{T}}x_i + b) \geq 1-\xi_i \\ \xi_i \geq 0 \\ i = 1,2,\cdots k \end{cases} \tag{8}$$

$$\min_{\alpha_i} \frac{1}{2}\sum_{i,j=1}^{k}\alpha_i\alpha_j y_i y_j \mathrm{K}(x_i,x_j) - \sum_{i=1}^{k}\alpha_i \quad s.t.\begin{cases} \sum_{i=1}^{k}\alpha_i y_i = 0 \\ 0 \leq \alpha_i \leq \mathrm{C} \\ i = 1,2,\cdots k \end{cases} \tag{9}$$

Where $\mathrm{K}(x_i,x_j)$, a kind of kernel function, used in linearly inseparable cases to solve the dimension explosion problem which generated from mapping dataset to high-dimension space. The optimal hyperplane can seek according to find α_{opt}.

III. Experiments

The experiments used 0, 1, and 2 as identification respectively to represent three data label types. And 837 samples were used as training data, the rest as testing data. Experiments were carried out in two hierarchies, the first layer distinguished between healthy and diseased, and the second distinguished benign lesions or malignant tumors, then two layers were combined to obtain three-class of classification results.

After pre-processing the clinical dataset, GBDT was used to select features, and classification algorithms were used to train models. Two layers both used the feature selection to reduce the dimensions of the dataset from 30 to 4 and 7 respectively. Finally, the models predicted and evaluated testing dataset.

Comparison experiments was used to illustrate whether feature selection can assist the method. And, different classifiers were used to show that feature selection is not only effective for a particular algorithm. The experimental process is shown in Fig. 1.

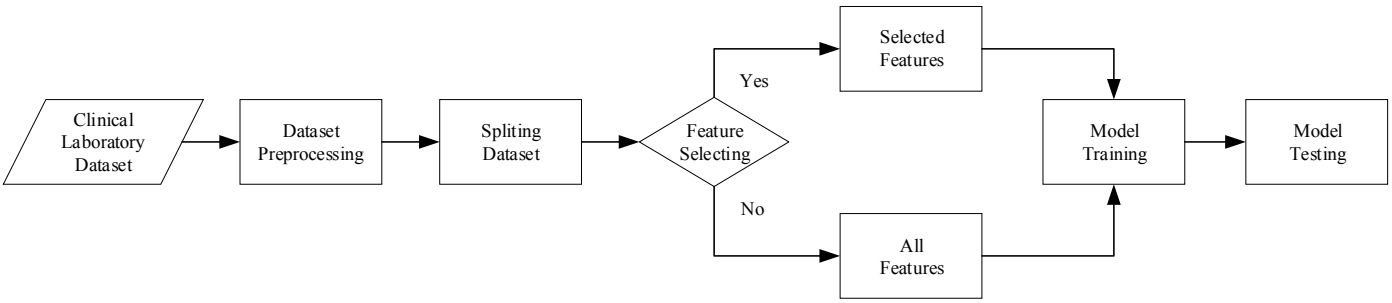

Fig. 1 The Flow of the Experiments

A. Evaluation Index

In the view of ordinary people, they hope the result of detection is consistent with real as much as possible, that is called accuracy. Clinical doctor often uses sensitivity and specificity to evaluate experiment results, which are based on confusion matrix. The label of real result (gold standard) is offered from biopsy.

Kappa index and accuracy, which based on three-class confusion matrices, used to evaluate the classification results. Kappa index is between 0 and 1. The closer to 1 it is, the higher consistency of results, higher reliability of forecasting and better results of classification it has. The accuracy is the ratio of correctly predicted samples in the total sample, higher accuracy indicates the model predicts more samples correctly and the result is better. The three-class confusion matrix is shown in TABLE I. The equation (10) calculates Kappa index, where p_0 is (11), p_e is (12), and the accuracy is equal to p_0.

TABLE I. Three-class Confusion Matrix

Prediction Result	Real Result			Total
	0	*1*	*2*	
0	n_{00}	n_{01}	n_{02}	$n_{0.}$
1	n_{10}	n_{11}	n_{12}	$n_{1.}$
2	n_{20}	n_{21}	n_{22}	$n_{2.}$
Total	$n_{.0}$	$n_{.1}$	$n_{.2}$	N

$$\text{Kappa} = \frac{p_0 - p_e}{1 - p_e} \tag{10}$$

$$p_0 = \frac{n_{00} + n_{11} + n_{22}}{N} \tag{11}$$

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$$p_e = \frac{n_{0.}*n_{.0} + n_{1.}*n_{.1} + n_{2.}*n_{.2}}{N*N} \quad (12)$$

B. Results

The results of experiments are shown in Fig. 2 and Fig. 3 respectively. Kappa index and accuracy are perfect whether the feature selection is used or not. Their fluctuation range is almost within 0.05. And the utilization of $\overline{X} \pm S$ represents the statistical values of Kappa index and accuracy, where $\overline{X}$ is average and S is standard deviation.

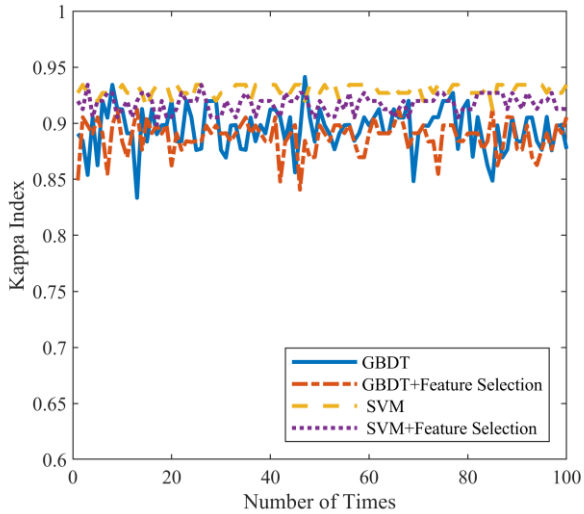

Fig. 2 Kappa index of model testing

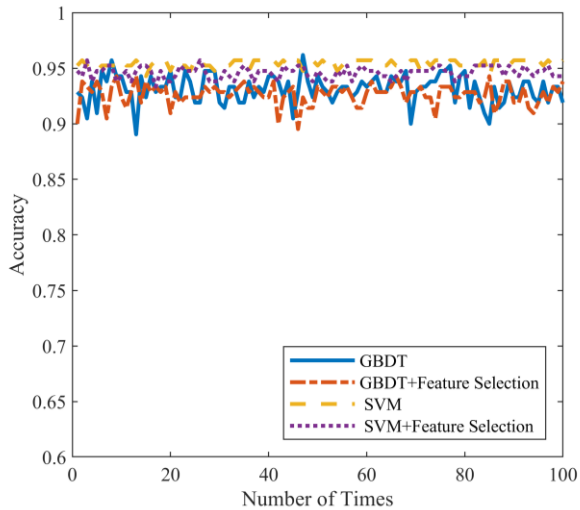

Fig. 3 Accuracy of model testing

TABLE II. COMPARISON OF EVALUATION INDEX

Model		Kappa	Accuracy
Classification Algorithm	***Feature Selection***		
GBDT	No	0.8951±0.0193	0.9313±0.0128
	Yes	0.8879±0.0144	0.9265±0.0096
SVM	No	0.9297±0.0053	0.9541±0.0035
	Yes	0.9175±0.0080	0.9460±0.0052

The results are shown in TABLE II. The indicators of evaluation model are a little different when using two classification algorithms and using feature selection or not. Besides, the result of SVM is a little better than GBDT.

IV. CONCLUSION

Early prediction and prevention of PLC is the most effective way to improve patient survival rate. To reduce the calculation cost while keeping classification performance, feature selection of GBDT was used before classifications of GBDT and SVM. The experiments show that the prediction results using the selected subset is almost identical to the results obtained by using all features. The Kappa index reaches the almost perfect level, and the accuracy is over 90%. Therefore, the proposed method can be applied to early screening of liver cancer. With the accumulation of the laboratory dataset, deep neural networks will be tried to do it in the future.

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